

Expectations Vs Reality of Conducting Ethnographic Research in Nigeria to Inform Autonomous Ground Vehicles Design

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ABSTRACT

We present a comparative reflection of our experiences designing and conducting ethnographic user research in understudied real-world contexts – Nigerian road traffic. We present our experiences planning and doing fieldwork to investigate and map Nigerian road users’ on-road experiences and perspectives on trust and safety in real-world traffic, towards identifying design factors to inform trustworthy autonomous ground vehicle design. We compare our expectations and plans for the fieldwork to the reality of conducting the research in a multicultural country like Nigeria. We describe how some contextual research factors – including geopolitical, institutional, cultural, infrastructural, safety, and trust factors – affected the fieldwork, and how we addressed them by adapting the methodology to be suitable for the research contexts, populations, and societies. Our insights may be useful for researchers designing or conducting ethnographic research in multicultural communities to capture understudied perspectives to inform technology design practices in a culturally sensitive manner.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • HCI design and evaluation methods; • Field studies;

KEYWORDS

Ethnography, User studies, Nigeria, Methodology, Contextual research design

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1 INTRODUCTION AND BACKGROUND

Ethnographic fieldwork is an enriching way to explore understudied research contexts; it enables researchers to probe people’s experiences, feelings, contexts, challenges, and needs. However, such fieldwork comes with known and unknown challenges and risks that must be continually managed by researchers. Drawing on existing literature and desk research conducted in Australia, we planned a 7-month ethnographic fieldwork in Nigeria to explore Nigerians’ perspectives on trustworthy autonomous ground vehicles (AGV) design factors. Despite our extensive planning, we faced methodological challenges and a significant gap between our expectations and the realities of doing the research in Nigeria. In this paper, we reflect on our experiences planning the fieldwork from Australia and then doing it in Nigeria, the challenges and factors that emerged, and the lessons we learnt that are likely to be useful for researchers exploring technology design factors in understudied multicultural regions.

The specific findings we obtained from this research are out of the scope of this paper. We focus instead on reflecting on the methodological insights gained from experiences in the field. To provide useful context for the fieldwork and the lessons learnt, we begin this paper with a description of the research project.

1.1 The duality of autonomous ground vehicles potential in societies

The integration of artificial intelligence (AI) systems in societies has manifested their potential to perform poorly or cause harm in uncontrolled real-world situations. These failures are especially prevalent when AI systems are not designed for the population and situations where they are being used [5, 19, 30]. One of the proposed ways to address this is to design AI systems to be trustworthy. By trustworthy design, we mean that trust is established in their design, development, deployment, and use [8]. To explore trustworthy AI design, we focus on AGVs – vehicles equipped with AI features so that they can perform driving functions [33] – as an instance of AI systems used in safety-critical contexts. AGVs’ usage on roads to assist human drivers’ or drive independently is anticipated to improve road safety by reducing human errors and road accidents. However, the integration of AI in AGVs also increases their performance uncertainty: AGVs have observably performed unsafely when used in traffic situations they are unfamiliar with [30] and made avoidable mistakes [34] due to their inadequate knowledge.

Driving norms and road users’ behaviours differ in various regions and countries [29]. For example, a zebra crossing is a safe place for pedestrians to cross the road in many countries, but not in Nigeria – even though the law says it should be safe [35]. So,

what safety means and how safety is ensured on road are localised. Usually, the locals understand the local traffic rules and norms, communication cues, and behavioural expectations in their road traffic, and act accordingly to ensure safety on road. This is because road traffics are safety-critical: they are highly sensitive contexts where mistakes from road users (and AGVs) can cause accidents. So, for AGVs to improve road safety as hoped, their driving automation need to be intelligent locally and must operate safely in their deployed real-world traffic. To ensure this, we need to understand the road systems AGVs will be used in, including the traffic rules, driving norms, and local safety mechanisms.

1.2 Nigeria and Africa – why we need to consider these perspectives and societies in AGV research

Africa could benefit from AGVs' potential (from a safety perspective), as a quarter of the global road fatalities are recorded in Africa even though only 2% of the global vehicles fleet is in Africa [36]. However, Africa is understudied in AGV research [2, 18] and AGV research is not representative of their potential global use cases in real-life road traffic. Even though automobiles are used globally in differing road traffic, AGV trust research has mostly originated from and focused on a few geographical regions, and primarily been done in controlled contexts like on mapped roads or factory grounds [4, 32, 41]. Despite this, cars are being integrated with driving automation like braking assistance, and sold by manufacturers to customers globally, including Africa [28, 31]. For example, 98% of the vehicles used in Nigeria are from the EU, USA, and urban Asia [40], where AGVs are predominately being developed; vehicles from these regions are now integrating driving automation [15, 31, 42].

African road traffic features diverse landscapes, practices, and road infrastructures that introduce challenges in AGVs and their AI designs [22]. Exporting or deploying AGVs to African countries like Nigeria with substantial variations in traffic situations, regulations, and norms [29] relative to the places they were designed and tested in, with no or flawed understanding of the local tacit and explicit road rules guiding the road users, can potentially yield unanticipated failures or issues in AGV performance and lead to accidents [see: [30]]. Our research aims to contribute a Nigerian perspective to trustworthy AGV research and to surface issues or factors to inform future AGV research. We focus on Nigeria as an example of an African nation with a high road crash rate [26], complicated and diverse road systems that may benefit from AGVs if AGVs are properly designed for their traffic, and likely to experience avoidable failures and harm from AGVs designed elsewhere. Nigeria is divided into 36 states, each with its unique driving, communication, and behavioural norms that must be considered in trustworthy AGV design for Nigeria.

1.3 Defining Trust – an example of how biases are embedded and extrapolated in technologies

We consider trust as a value with ethical and safety implications that is often designed for in AI systems [12]. Situational, societal,

cultural, and personal factors [16, 23] can shape how trust is experienced and defined – as can the actors involved. To research trustworthy AI and AGV design for Nigeria, we first explored trust definitions of relevance to humans and automation interaction research. We found that many trust definitions have been proposed, as summarised in [37]. As a starting point, we adopted the definitions of trust from Lee and See [21] and Friedman et al. [12]. Lee and See explored trust as a factor that facilitates human-automation collaboration, while Friedman et al. investigated it as a human value influenced by our ethical and moral views that are usually designed for in technologies. We adopted both definitions to enable us to investigate trust as a human-automation interaction factor as well as a human value that facilitates humans' interactions with others – shaping their mindsets and expectations.

In addition, we explored the Nigerian and African definitions of trust. We found that trust is understudied in Africa and Nigeria [9], and trust research involving African populations usually draws on prevalent definitions from Western literature. Values represent what people deem important [12]. As noted in [12, 39], technologies and their infrastructure encode the values of those who shaped them. So, not including Africans' perspectives and values in AI systems research inherently neglects African values and cultures in AI design and exposes Africans to potential harm from such systems [1, 5, 43]. To integrate African values into our research, we explored how indigenous African languages describe trust. We found that trust is usually referred to in African words that are synonymous with faith, expectations, hope, or dependability [17]. In the next three sections, we introduce the project's contexts and research questions, research methodology, and key elements involved in planning and doing the fieldwork in order to provide context for the insights, lessons and key takeaways that follow.

2 CASE STUDY: EXPLORING ROAD USERS' EXPERIENCES IN THE NIGERIAN ROAD TRANSPORT SYSTEM

2.1 The project team and context

The research was designed to answer the question: *how do we design AGVs that are worthy of being trusted and used in African societies using Nigeria as a case study?* Although Nigeria is the primary research context, the research is being conducted as a PhD project at a university in Australia. The primary investigator (PI) is a Nigerian PhD candidate whose background is in cybernetics and computer science. The PI, as a Nigerian, had an insider perspective on the research context and population that shaped the research design. She and the PhD research committee brought an outsider perspective to the research as follows: (1) we designed the research remotely, in a foreign institution; (2) the institutional policies that shaped the research are more representative of Australia's values, research factors, and realities than Nigeria's; (3) the literature we relied on are Western and integrate Western assumptions; and (4) during the research planning, the PI had not lived in Nigeria for 3 years. Hence, we designed the research to be responsive and grounded in the research contexts.

3 OUR METHODOLOGY AND APPROACH

The research was exploratory, pragmatic, and used a multi-method approach - to enable us to comprehensively understand the currently understudied research problem [7, 14]. To explore the problem at a systems level, we drew on second-order cybernetics. Cybernetics considers humans and autonomous systems to be goal-driven [38]. This goal-driven lens can help in understanding humans' and machines' behaviours [10] and in designing systems that integrate or interact with heterogeneous agents. Taking on this perspective, we viewed Nigerian road users as purposeful, social, and values-driven agents. We also viewed road traffic as systems consisting of diverse actors who engage in multiple and, at times, overlapping interactions that affect their environments and surrounding road actors. Hence, to design AGV interfaces and interactions that fit Nigerian road systems, we designed a field study to understand how people and technologies currently communicate and interact in road traffic.

We explored the current road users' experiences to anticipate how Nigerian road users might interact with AGVs, including: what AGVs need to communicate, how AGVs may react or interpret road users' signals appropriately, etc. Hence, the study explored road users' actions, interactions, perceptions, and expectations on roads. We used a multi-method approach to allow us to capture and triangulate varied contextual data, safely include diverse perspectives, manage the risks on road appropriately, and respond to the risks that emerged on the field without significant setbacks. The sampling methods we adopted include: convenience, snowball, and stratified purposeful sampling [6]. We chose these sampling methods because they are appropriate for exploratory research, and they also serve to reduce some of the risks in recruiting participants, such as not having access to the participants or recruiting uncooperative or untrusting participants. Findings from this exploratory research were not intended to be generalised. Rather, we aimed to surface and highlight research areas for further study.

3.1 Methods

The primary methods used were ethnographic and qualitative methods. We adopted qualitative methods to capture the participants' experiences and perspectives in their own languages as much as we could, with the aim of creating a rich understanding of what people do and why they do them. We did this to identify the goals, values or belief systems that are associated with people's actions and interactions as well as identify new road users' patterns and behaviours [20:5] of relevance to AGV design. The data collection methods used were as follows:

- **Autoethnography.** The primary participant of this method was the PI. The PI conducted autoethnography on roads by immersing herself in Nigerian road traffic as a road user, taking on different roles - driver, pedestrian, and passenger. With autoethnography, the PI was able to capture, reflect on, and present her first-hand experience of on-road communication, interactions, and norms, and explore some of the participants' claims. Data were captured as reflexive journaling, video recordings and pictures of her experiences.
- **Contextual inquiry.** This method was used on-road on pedestrians, and on passengers and drivers in cars. The PI used contextual inquiry to gain second-hand experiences of road interactions, communication, and culture by observing and interviewing road users about their experiences while navigating the road traffic. It was an immersive way to learn about their perspectives, feelings, experiences, and the road contexts.
- **Interviews.** The participants of this method were motorcyclists, hawkers (people who sell on the streets or on highways), Nigerian road transport experts, and automobility experts. This method was used to (1) capture the perspectives of the road user groups we could not safely use contextual inquiry for, and (2) conduct interviews of AGV experts, Nigerian automotive manufacturers, and some road transport government agencies. The PI used semi-structured interviews for planned interviews and unstructured interviews for the unexpected ones.
- **Observation.** This method was used to observe the actions (including associated motivations, perceptions, and information exchange) and the resulting reactions from the perspectives of various road user groups in differing real-world road traffic situations. The PI also observed the road infrastructure, technologies, and vehicle roadworthiness tests conducted at a Nigerian vehicle inspection centre.

4 PLANNING AND DOING THE FIELD RESEARCH

Drawing on ethnographic literature, we planned the fieldwork remotely based on known factors. In this section, we reflect on the key research factors we planned for while in Australia and how they played out in Nigeria.

4.1 Locations and overall duration of the fieldtrip

To enable a comparison of the varying road norms and expectations in different Nigerian states, we chose to do the fieldwork in two major Nigerian cities, Lagos and Abuja. We chose them as examples of Nigerian states with varying road norms, road architectures and infrastructure, and multicultural residents. To drive safely in any of these states, drivers must understand the road's cultural contexts and norms. Influenced by the significant cost of travel between Australia and Nigeria, we decided that the fieldwork would involve a single trip to Nigeria, for 7 months. This decision was possible because the PI is originally from Nigeria and did not need to learn the local languages and customs on arrival. We estimated that 7 months would give the PI ample time to resettle in Nigeria, scope the research contexts, and adjust the plan to fit the realities in Nigeria. It also left buffer time for the PI to follow up with participants or include new stakeholders that might emerge from the participant sampling while the PI was in Nigeria. From the PI's prior experience, she correctly anticipated that it would be easier to follow up in person as some participants may not have internet access or be inaccessible online.

4.2 Researching and defining trust from Nigerians' perspectives

To capture the Nigerians' definitions of trust and its factors, we made a starting list of trust elements to be investigated by extracting key elements from Friedman et al. and Lee and See definitions as well as the African terms for trust. Hence, we explored participants' experiences of dependability, vulnerability, faith, hope, benevolence, and betrayal in situations that they considered risky and costly. We iteratively updated this list in the field to reflect the participants' languages for trust from gathering data on trust as (1) described in participants' indigenous language; and (2) experienced on-road and off-road.

We planned to hire a local research assistant to converse with participants in some of their indigenous languages so that we could (1) include participants who are not English literates or cannot converse in the languages the PI speak; (2) capture Nigerians' trust definitions in their languages. For context, Nigeria has about 512 indigenous languages [44], and the PI speaks only 3 of them fluently. Through the research assistant, we wanted to include the 3 major languages spoken by many in Nigeria – Igbo, Yoruba, and Hausa, and the Nigerian Pidgin English.

4.3 Defining the sample size and target participants – a research boundary judgement

Defining the sample size was a research boundary decision that depended on another factor: the target participant groups. The target groups implicitly dictated whose voice we included, prioritised, or excluded. The road transport system permeates society and affects everyone, but we cannot include every Nigerian road stakeholder's voice, given the scope of our research. Hence, based on considerations of the research question and goals, as well as people's abilities to participate safely or give informed consent, we defined the target participants groups as:

- Group 1: Nigerian road users [13] - adult motorists (passengers, private, commercial, and ride-hailing drivers), pedestrians, cyclists, motorcyclists, hawkers, and traders.
- Group 2: Nigerian and Lagos governments road transport agencies, Nigerian automobile manufacturers, and autonomous vehicles experts in Nigeria and beyond.

Considering the diversity in each participant group and the likely research risks like inability to access research sites or participants, we defined the sample size as a range: 20-40 people. This was to enable us to be responsive and adjust the research schedule and sample size, or revise the target participants group as needed in Nigeria; e.g. we removed cyclists from the target groups because the PI could not access them as many Nigerians do not cycle [24]. This also reflects the limitation of the sampling methods. Stakeholder analysis was continuously done in the field to update the identified stakeholders and participant lists as the PI engaged with Nigerians and the road system. The new stakeholders added to the list were: the Lagos road transport planning departments, Nigerian road safety agencies, traffic police, the UNDP, driving educators, insurers, delivery riders, and AI researchers.

4.4 Participant recruitment and recruitment channels

One of our research goals was to be inclusive. Therefore, we wanted the recruitment channels to be accessible to the Nigerian public and the target participants. The recruitment channels we considered were: emails, messenger apps, social media, and person-person introductions. The PI knew that while some Nigerians may not be active on email or social media, many Nigerians are accessible digitally via instant messengers or SMS. So, we planned to use email, SMS, instant messaging platforms like WhatsApp, remote or in-person introductory meetings, or calls organised by people in the PI's network, as initial contact channels. While in Nigeria, the PI maintained contact with the participants using any of these channels depending on the participants' stated preferences after initial contact.

4.5 Defining the participation remuneration in range

Deciding the appropriate remuneration amount for participation was challenging in the institutional ethics application process because the PI considered the equal remuneration recommended by the ethics committee as unfair due to the diversity of the participants, the varying associated costs of their participation, and the varying cultural appropriateness of remunerating various participants. Therefore, the PI had to argue against equal remuneration. To ensure the remuneration was fair and culturally sensitive, we determined a range of amounts that were ethical and appropriate to pay participants, drawing on the cost of living in Nigeria. The remuneration was in different forms. This was to enable us to be culturally sensitive to the people for whom receiving monetary reimbursement may be considered disrespectful or inappropriate. Some of the remuneration options were – money, gifts, vouchers, fuel or transport fare, etc., costing within the set remuneration range.

5 INSIGHTS – RESEARCH FACTORS THAT EMERGED DURING THE FIELDWORK IN NIGERIA

5.1 Institutional factors – how the research design embedded institutional requirements and assumptions

We refer to institutional factors as the required policies and procedures for doing research on humans or overseas in our research institution and their assumptions that affected the research. Examples of these are described below.

5.1.1 The applicability of institutionalised policies on risk assessment – by who? And for whom. Foregrounding the safety of the participants and researchers is a key element of human research ethics. Human ethics committees assess research proposals to support this. For our research ethics application, we drew on *Australia's National Statement on Ethical Conduct in Human Research* [27], *Australia's DFAT Smart Traveller travel advice* [3], *institutional WHS policies*, *Australia's COVID-19 health advice*, road safety reports from Nigeria [25]; previous research done in real-world traffic e.g.

[17], the PI's knowledge of the Nigerian road transport system, and other relevant sources to identify and mitigate the potential physical, cultural, and mental risks of conducting the fieldwork on and off roads.

Through the ethics process, we observed a significant disconnect between the Australian-based institutional and the PI's Nigerian views of the risks identified and their likelihood in Nigeria. For instance, our institution relied on the DFAT for travel advice; the DFAT rated Nigeria as a high-risk destination [3]. They highlighted diseases, theft, civil unrest and terrorism, harassment, sexual abuse, etc., as travel risks. The PI contested this "overestimation" of the risks given her 25 years of experience in Nigeria without experiencing these. DFAT risks "reflect the risk for average Australian travellers to this country." [3] Hence, the travel advice felt overestimated for the PI, who is from Nigeria. These conflicting views in the risk assessment reflect the mismatch in how locals and outsiders view or experience societal risks. Regardless, we used the DFAT rating as it was institutionalised with no means to adapt the risk ratings to reflect the PI's local insights. To get research approved, the PI had to assess and plan for risks from the lens of an Australian going to Nigeria for research, and not from the PI's perspective as a Nigerian. Adopting the institution's lens and designing the plan to comply with the policies as designed for Australians travelling to Nigeria for research enabled the fieldwork approval.

The PI conducted the 7-month research safely with no incident. Her experiences as a Nigerian researcher and resident who had lived safely in various regions in Nigeria for 25 years reflected her local knowledge of safety in Nigeria. The institution's policies and ethics committee failed to see how her past experiences as a local in Nigeria had equipped her with local strategies that effectively mitigated the risks that were identified for an average Australian visiting Nigeria.

5.1.2 The human ethics approval processes impact on the research timeline and inclusiveness. While our initial planned fieldwork duration was 7 months, the extended delays and questions that arose through the ethics approval process resulted in us condensing the fieldwork plan into just over 4 months and reducing the research contexts from three Nigerian cities, which was the initial plan, to two cities. We thought this would reduce the number of participants, but we engaged more - about 50 participants - due to the diversity of the road transport stakeholders. The delay also hindered us from hiring research assistants as the necessary considerations relating to research ethics, research assistant training, data privacy and security, etc., would have further delayed the research ethics approval process, which had already significantly impacted planned timelines and scope. As a result, the PI only included participants who could speak either English or any of the Indigenous languages the PI understood: Ebira, Nigerian Pidgin, Yoruba, and Hausa.

5.2 Evolution of relative perceptions of risk

The PI planned to reassess the risks in consultation with the locals once she arrived in Nigeria to adapt the research plan to be contextually safe. When the PI arrived in Lagos, everything on the roads initially seemed hazardous, and her initial risk assessments were higher than the locals'. For example, we initially wanted to do diary studies with hawkers, to which the consulted hawkers

agreed. The PI felt it was unsafe, given her fresh experience of Lagos traffic, but the hawkers disagreed with her. Over time, as the PI immersed herself in the road traffic, she noticed that her view of road users' languages and actions evolved. She later embodied the norms like a local when she stopped overthinking people's actions and navigated the roads instinctively. Soon, the PI could not see many of the risks she initially saw on the roads. This evolution made her realise that her initial views of the risks were clouded by the institutional Australian-based policies that she had immersed herself in just before her trip, with greater focus on worst-case scenarios. This also affected her initial research experience: she could not fully immerse herself in the road traffic or easily approach strangers for recruitment as they seemed unsafe.

5.3 Cultural factors – embracing the Nigerian flexible flow in research workflow

The PI went to Nigeria expecting that her interactions with participants would be as structured and procedural as observed in literature. However, most interviews did not play out that way. Many participants wanted to participate immediately when the PI had intended to recruit them or schedule interviews. The PI unexpectedly recruited many road users - mostly drivers - who wanted to participate instantly while she was on the road for reasons unrelated to the research. For instance, she had to take notes for an unplanned interview on a receipt as it was the only paper with her. For scheduled meetings, participants often arrived late or early due to several factors like prior unplanned appointments or traffic congestion. Hence, the PI was quickly reminded of the flexible lifestyle in Nigeria and adapted her approach to fit this by ensuring that she was always prepared for these scenarios, whether she was on road for research or not. She planned for the unexpected and expected by taking key research tools like ID card, consent forms, information sheet, audio recorder, notepad, and pen everywhere she went. Whenever she had scheduled multiple interviews in a day, she would have at least a 2-hour buffer period between the interviews to give allowances for lateness. Additionally, since most of the interviews were unanticipated and the PI did not have time to prepare the interview questions in advance, she learnt to "wing it" and conduct interviews without any preparations by being more conversational than formal. The PI also learnt that Nigerians were usually keen to participate instantly because she was a PhD student, and they wanted to help ensure her PhD was successful. Hence, the PI considered the PhD fieldwork as a communal endeavour that was greatly supported.

5.4 Participant recruitment in Nigeria – contacting participants

The PI learnt that recruiting employees of organisations to speak on behalf of their employers required seeking consent from the organisations and not employees. Initially, the PI could not contact Nigerian companies and government agencies via email; they never responded. But when she contacted them in person by hand-delivering mail to their physical offices, she got immediate responses and secured appointments easily. Also, many of the participant drivers were recruited and interviewed while on the job – driving – because it was harder to find drivers outside their work

environment, and it was more difficult for the drivers to find time to meet elsewhere for the interviews. Aside from the drivers and organisations, the initial contact methods for other participants were WhatsApp and introductions facilitated by the PI's mutual contacts. In Nigeria, the participants were generally more accessible and easier to recruit in person than remotely.

5.5 The relativity of participants' trust – and trustworthiness factors in the field

During recruitment, the PI observed varied trust behaviours from the participants. Recruits that had prior connections with the PI usually agreed to participate and sign the consent form without asking many questions – as if trust was already built. In contrast, the unfamiliar recruits were usually willing to participate and be audio-recorded until they were asked to sign the consent form. They initially hesitated and enquired about the research and why they had to sign the form as if they were assessing for risks and trustworthiness – because their signatures bind them and can expose them to potential harm. This distrustful attitude towards the consent form misaligns with its purpose – to protect participants. The participants felt that signing it made them vulnerable to harm if the terms were malicious. Since the PI could not spend an extended period living with the recruits to gain their trust as in many cultural ethnographies, she had to gain people's trust or confidence within minutes of meeting them.

5.5.1 Gaining strangers' trust in minutes on Nigerian roads. To help establish her trustworthiness instantly during recruitment, in addition to her personal experience, the PI consulted with her local contacts and friends who are entrepreneurs in Nigeria to adopt these trust mechanisms: appearing clean and smart-casual, having an elevator pitch, avoiding attitudes that create suspicions (not trying too hard to convince the recruits and giving them the time needed to assess the information sheet and consent form). The PI learnt that in building initial trust, the recruits assessed her through how she described the research and herself; her mindset was reflected in her attitude and in the participants' evaluation of her capabilities and trustworthiness. The PI observed that as she became more confident in her capabilities to conduct the fieldwork, she was able to earn participants' confidence and recruit them much faster. In addition, to make the participants feel safe, they picked the interview locations, which were either their homes, offices, or restaurants.

5.6 The cultural sensitivity of remuneration

The remuneration amount and form varied depending on the estimated time and financial cost of participation. The remuneration forms included airtime vouchers, gift cards, soft drinks, cash notes, and bank deposits. Participants' remuneration was contextual and culturally sensitive as the PI observed varied attitudes to the remuneration from various participants. The older and wealthy participants never accepted the remuneration because they participated to support the PI's research, given their direct or indirect relationships with the PI. In this context, no amount of remuneration could equate the value of their time and offering them remuneration was seen as putting a price on these, and hence, was insulting. Also, it was inappropriate to remunerate government officials. Instead, the

PI thanked them for their participation. For participants in lower income brackets, some preferred cash notes to gift cards because they lived far from the gift card stores. They also said that receiving cash notes gave them better utility for the remuneration than spending it shopping at the store.

5.7 Geopolitical factors – elections and cash policies

The Nigerian 2023 general elections were conducted during the fieldwork in Nigeria. This manifested some unanticipated research factors like cash scarcity. Shortly after the PI arrived in Nigeria and a month away from the elections, the Central bank of Nigeria announced the redesign of the Nigerian cash notes (Naira) and that the old notes would expire 3 months after the announcement – a few weeks until the elections – to prevent electoral fraud and to enable a cashless economy [11]. This created a temporary severe cash scarcity nationwide, which delayed research progress as Nigeria has a cash economy. It became difficult to access cash notes to pay participants, pay for transport, or other research costs where notes are the only payment method. The PI had to reschedule some interviews and research activities as participants could not access cash to pay for their transport fare or car fuel to get to the research sites – especially for contextual inquiries. So, in the first 3 months of participant engagement, participants who preferred cash were remunerated via bank transfers.

5.8 Infrastructural factors

Some infrastructural and societal factors in Nigeria that informed the research schedules were: electricity supply, traffic congestions, transportation access, etc. Some issues like unstable electricity supply were nationwide problems, while other issues like traffic congestion were local to some states. To manage the electricity supply issues, the PI only stayed in houses where constant power supply was assured – at additional cost. She also took backup power supplies – power bank and batteries – to the field to power the camera, sound recorder, and phone as needed. In Lagos, serious traffic congestion results in immediate high transport fare hikes, which also increased the research cost. To manage this, the PI planned her activities in the field around the road traffic peak hours. This limited her hours in the field in Lagos to 10 am – 2 pm and she could only do 1-2 interviews per working day. In contrast, she could be in the field all day in Abuja and do up to 4 interviews in a day. The Nigerian hot climate, traffic congestion, and other factors on the roads/sites made conducting the fieldwork draining and slower than anticipated. As a result, the PI learnt to plan for rest time.

6 LESSONS AND KEY TAKEAWAYS

Doing fieldwork overseas in Nigeria – a multicultural and multilingual country – to explore AGV design factors created insights on the mismatch between our expectations and realities of doing culturally sensitive ethnography in real-world safety-critical environments towards the design of AI systems. The realities of the mismatch and their resulting challenges emerged in various ways. Some of these were: (1) aligning institutional research policies with a fieldwork plan for an overseas context and population; (2)

managing uncertainties stemming from the fluidity of Nigerians' governments' policies, plans, and traffic situations; (3) the difficulty of setting boundaries for research that affects society at large; and (4) navigating the sensitivity of recruiting, remunerating, and building trust with diverse participants. In addition, the fieldwork was significantly affected and sometimes delayed by external factors that were either global (e.g. Covid-19), infrastructural (e.g. traffic congestion), geopolitical (e.g. elections), or institutional (e.g. travel policies). Doubling the estimated time of our plan and taking a contextual approach to the research design by leaving low-level details about the plan execution to when the PI arrived in Nigeria enabled us to be responsive to the unanticipated challenges that emerged and successfully conclude the fieldwork on time.

We found that designing ethnographic research and assessing its risks are best done in the field – to avoid overestimation or underestimation. This enables consulting the locals on the fieldwork design and risk assessments, which enhanced our research plans' safety and robustness. We also learnt that the expectations and assumptions underpinning fieldwork design may be flawed, and so, researchers should be critical about assessing their assumptions and be open to letting the fieldwork realities shape the research plans and execution, especially when doing research in uncontrolled social environments where unexpected issues may emerge. For the PI, being contextual and responsive in her approach helped her to be prepared for the unexpected. For instance, she prepared for as much as imaginable by always carrying the essential tools everywhere while in Nigeria.

In conducting the research in Nigeria, where African values like communalism, respect for elders, etc., are significant in people's relationships and interactions, the PI found that being aware of the cultural norms and integrating them into the research helped ensure that the participants and their societies were respected in the fieldwork. She found that listening to the participants, especially when the researcher's views on safety, rights or fairness conflicted with the participants', made the participants feel safe and respected. Appropriate remuneration was contextual and varied: there was no one solution to ensuring fair remuneration off-site or on-site as the research participants were diverse and cut across various socioeconomic classes.

7 NEXT STEPS

The PI is now analysing the data collected using thematic analysis, stakeholder analysis, and systems mapping to understand and map Nigerian road traffic, the road users' lived experiences on road, the road norms or laws, and trust-associated values. These will be used to define AGV design requirements for Nigeria that will help ensure AGVs and their AI are designed to be safe, inclusive, and trustworthy in Nigeria through consideration of Nigerians' road users' dynamics, values, welfare and environments in their design and standardisation. Although the research is exploratory and was only conducted in 2 Nigerian cities, the methodological insights are likely to be beneficial to researchers doing user research on understudied or multicultural perspectives in a culturally sensitive manner.

We encourage more technology design research to foreground the values, lived experiences, and welfare of all stakeholders, especially in Africa, to enable them to identify novel design factors and insights that are valuable to diverse user groups. We recommend that (1) ethnographic user research with foreign or understudied populations include and prioritise the participants' or locals' opinions and values; (2) researchers should continuously reflect on and critique the assumptions underpinning the resources that shape the research, including: institutional policies, existing literature, or methodologies; (3) researchers should adapt and contextualise research resources as needed in consultation with the participants to fit their contexts, realities, and values; and (4) researchers should ensure they reflect on how the research boundaries judgement may limit the applicability of its outputs. Institutions should also critique who their policies protect and if necessary, redesign them to be more inclusive.

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